Portfolio Optimization in R

Mean-Variance Optimization vs Equal Weight vs CVXR Approach

Portfolio optimization is a fundamental concept in financial engineering that focuses on allocating investments across multiple assets to achieve an optimal balance between risk and return. The theoretical foundation of this approach lies in **Modern Portfolio Theory**, introduced by **Harry Markowitz**. This framework revolutionized investment decision-making by quantifying the trade-off between expected return and risk using statistical measures such as mean and variance.

In this report, we explore three approaches to portfolio construction:

1. Mean-Variance Optimization (MVO)
2. Equal-Weighted (EW) Portfolio
3. Convex Optimization using CVXR

The objective is to compare these methods in terms of efficiency, practicality, and flexibility.

Data and Preprocessing

The analysis is conducted using historical financial data of three major technology companies: Apple (AAPL), Microsoft (MSFT), and Google (GOOG). Daily stock prices are converted into returns, which serve as the primary input for portfolio construction. Preprocessing steps include handling missing values and organizing the data into a consistent format suitable for analysis.

Returns data is essential because it captures the relative change in asset prices and allows for statistical modelling of risk and return characteristics.

Methodology

1. Mean-Variance Optimization (MVO)

Mean-Variance Optimization aims to construct a portfolio that either maximizes expected return for a given level of risk or minimizes risk for a given expected return. This is achieved by solving an optimization problem that considers:

- Expected returns of assets
- Variance and covariance among assets The optimization is subject to constraints such as full investment (all capital is allocated) and non-negative weights (no short selling). The resulting portfolio lies on the efficient frontier, representing the most efficient combinations of risk and return.

2. Equal-Weighted Portfolio (EW)

The Equal-Weighted portfolio is the simplest form of asset allocation, where each asset is assigned the same weight regardless of its risk or return characteristics. This approach does not rely on statistical estimation or optimization techniques.

Despite its simplicity, the EW portfolio is widely used as a benchmark due to its robustness and ease of implementation. It avoids estimation errors that may arise in more complex models but may not achieve optimal performance.

3. Convex Optimization using CVXR

Convex optimization provides a mathematically rigorous approach to solving portfolio optimization problems. Using the **CVXR** package, the portfolio variance is minimized subject to linear constraints such as full investment and non-negativity of weights.

This approach directly formulates the optimization problem using matrix algebra and solves it using efficient numerical solvers. It allows for greater flexibility in incorporating additional constraints and objectives, making it suitable for advanced portfolio design.

Results and Analysis

1. Portfolio Allocation

The Mean-Variance Optimized portfolio assigns weights based on the relative performance and risk characteristics of each asset. Assets with higher expected returns and lower volatility tend to receive higher allocations.

In contrast, the Equal-Weighted portfolio distributes capital evenly, ignoring asset-specific information. The CVXR-based portfolio produces allocations similar to MVO, as both methods aim to minimize variance under similar constraints.

2. Risk and Return Comparison

The comparison of portfolio performance reveals clear differences among the approaches:

- The MVO portfolio typically achieves lower risk for a given level of return, demonstrating its efficiency.
- The EW portfolio provides moderate performance but lacks optimization, resulting in suboptimal risk-return trade-offs.
- The CVXR portfolio closely aligns with theoretical expectations and often mirrors the results of MVO due to its precise mathematical formulation.

3. Key Observations

- Optimization-based portfolios outperform naïve allocation strategies in most scenarios.
- Equal weighting, while simple, can still perform reasonably well in uncertain or volatile markets.
- Convex optimization offers greater control and flexibility, enabling the inclusion of additional constraints such as transaction costs or diversification limits.

Advantages

- Mean-Variance Optimization provides a systematic framework for decision-making.
- Convex optimization ensures mathematically optimal solutions.
- Equal weighting is easy to implement and requires no assumptions about market behaviour.

Limitations

- MVO is highly sensitive to estimation errors in expected returns and covariance.
- All methods assume stable market conditions, which may not hold in reality.
- Transaction costs, taxes, and liquidity constraints are not considered in this analysis.

Conclusion

This study demonstrates that optimization techniques play a crucial role in constructing efficient investment portfolios. The Mean-Variance Optimization approach, implemented using **PortfolioAnalytics**, provides superior risk-return performance compared to Equal Weighting. Meanwhile, convex optimization using CVXR offers a flexible and robust alternative for solving portfolio problems.

Although the Equal-Weighted portfolio remains a useful baseline, advanced optimization methods are essential for achieving better financial outcomes in modern investment environments.

7. Future Work

Future improvements to this study may include:

- Incorporating additional asset classes such as bonds or commodities.
- Modelling transaction costs and market frictions.

- Implementing dynamic portfolio rebalancing strategies.
- Visualizing the efficient frontier for deeper insights.

8. References

- Markowitz, H. (1952). Portfolio Selection
- Documentation for PortfolioAnalytics
- Documentation for CVXR